EPITOPE

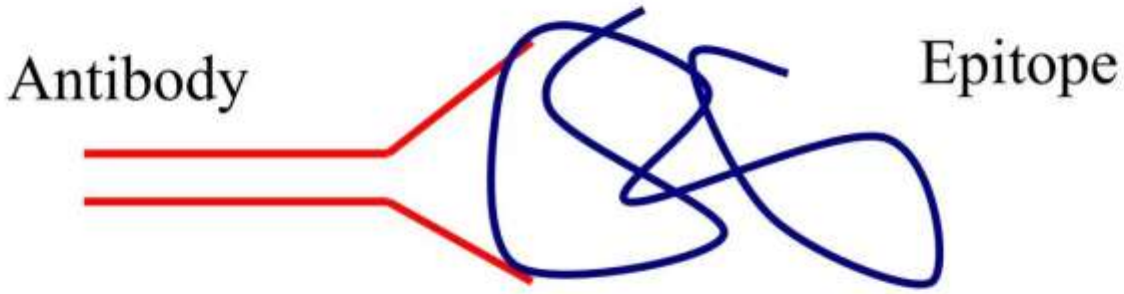
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Epitope

Antigenic determinants or Epitopes are the portions of the antigen molecules which are responsible for specificity of the antigens in antigen-antibody (Ag-Ab) reactions and that combine with the antigen binding site of Ab, to which they are complementary.



Epitope's Properties and its

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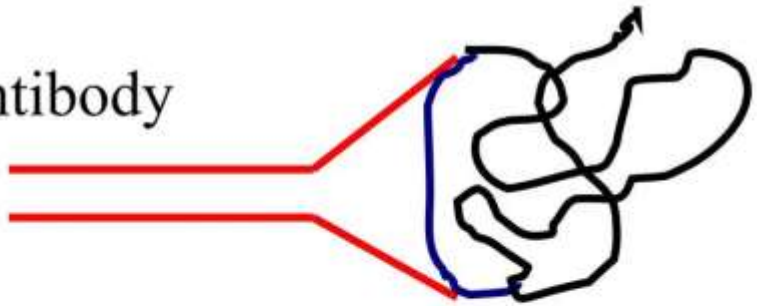
They occur on the **surface** of the protein and are more **flexible** than the rest of the protein

They have high degree of **exposure to the solvent**.

The amino acids making the epitope are usually **charged** and **hydrophilic**

Sequential / Continuous Epitopes

Antibody



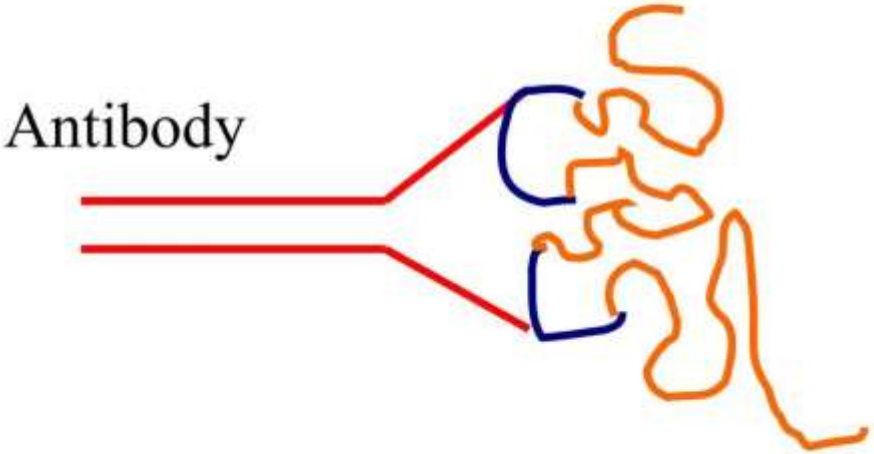
Recognized by T_H cells

Linear peptide fragments

Amphipathic helical 9-12 mer

Paratope

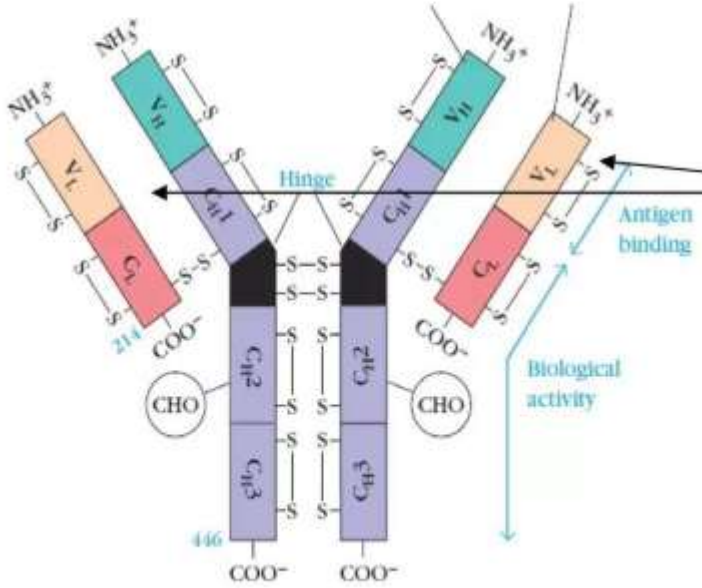
Conformational / Discontinuous Epitopes



Recognized by both T_H & B cells

Non-linear discrete amino acid sequences, come together due to folding

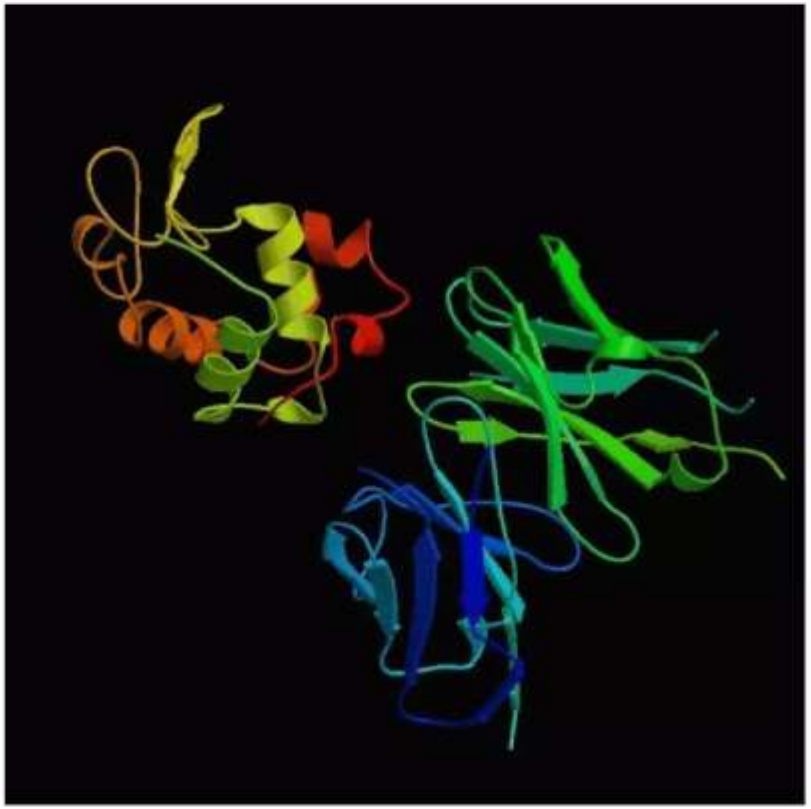
Exposed 15-22 mer



Paratope

Sites of Antigen binding on Antibody molecule

Immunological Processes

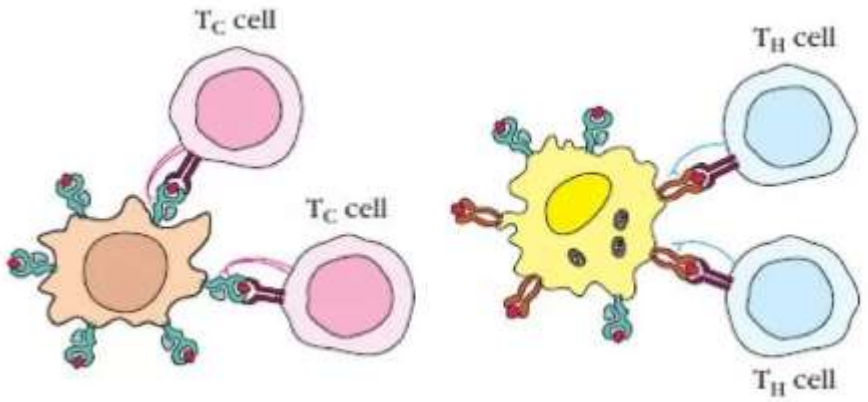


Ag-Ab Complex

MHC molecules function as antigen-recognition molecules

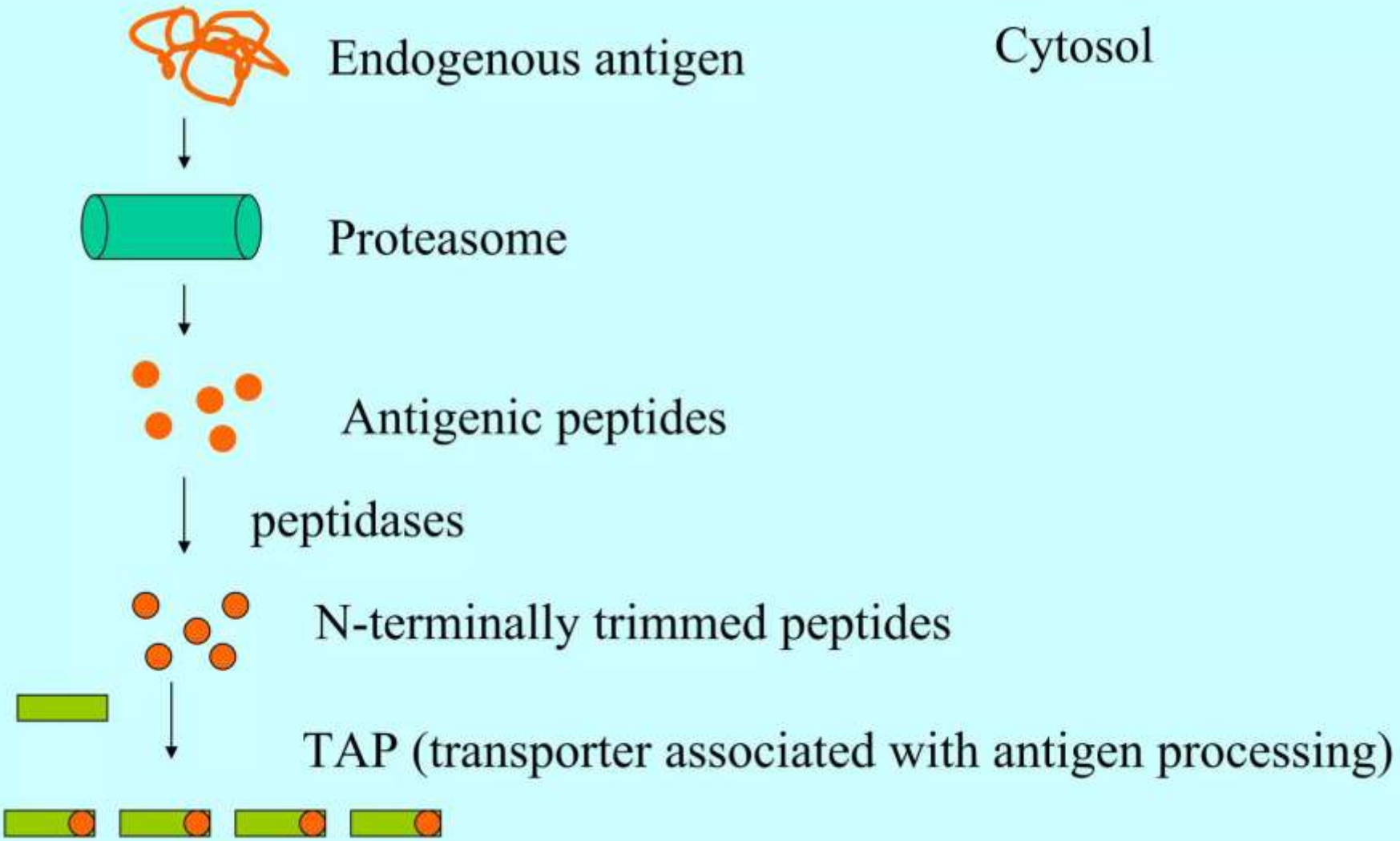
Class I – Presents Ag so that T_C cells recognize & kill. Requires CD8+ on T_c

Class II – Presents Ag so that T_H cells recognize & kill.

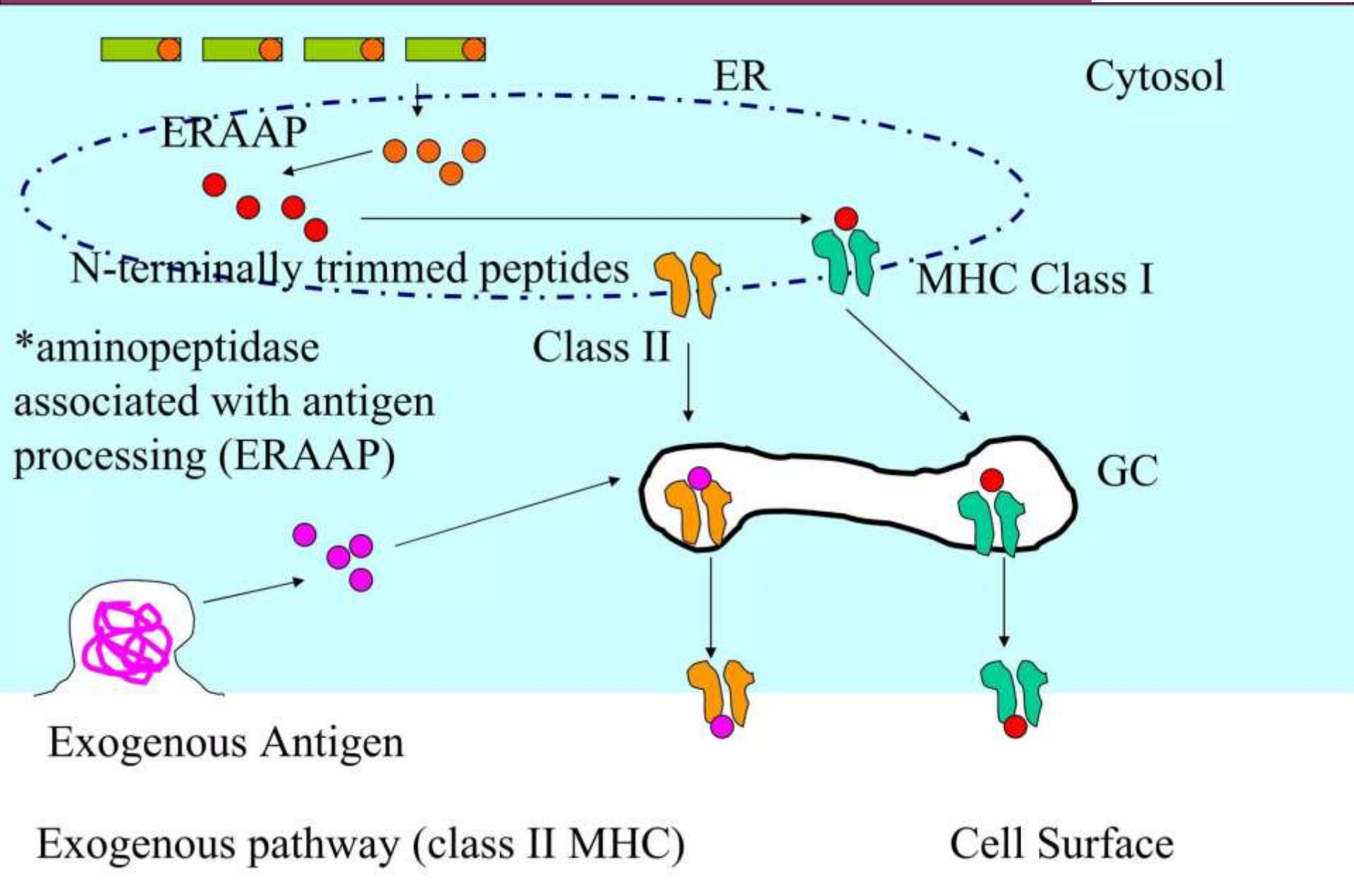


Immunological Processes

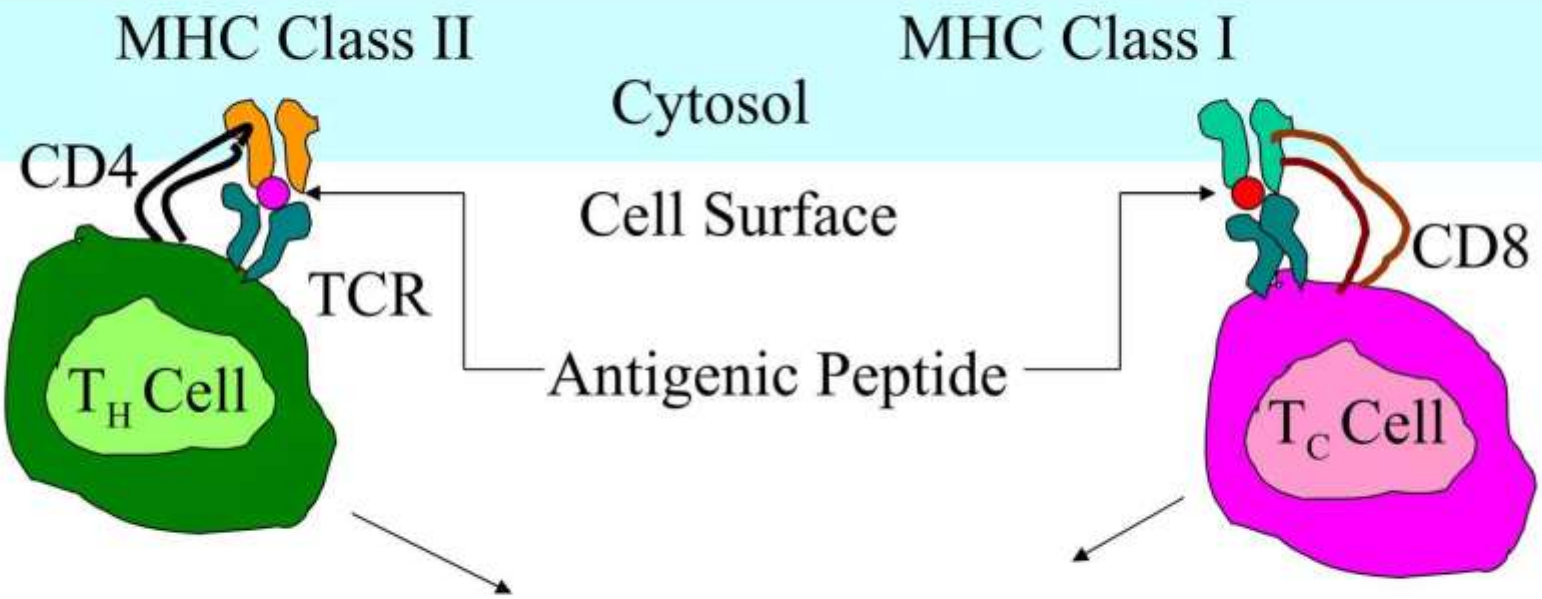
Endogenous pathway (class I MHC)



Immunological Processes



Immunological Processes



Immuological Responses

Where MHC are expressed ?

Class-I all nucleated cells e.g. virus infected cells

Class-II APCs (macrophages, B lymphocytes, and dendritic cells)

T_H cells expresses CD4,CD4 recognizes MHC class II molecules

T_C cells expresses CD8,CD8 recognizes MHC class I molecules

B-Cell Epitope Prediction

Hopp & Woods method

…….-Ser-Thr-Cys-Asn-Glu-……

…….-Ser-Thr-Val-Asn-Glu-……

1 + 2 + 10 + 4 + 5 = 22

e.g. ser-1, thr-2, cys-3,asn-4,glu-5, etc,…..x-10
alignment score = 22

Based on alignment score predict Antigenicity

B-Cell Epitope Prediction

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Welling's method

Database of Known Epitopes

↓

% of Epitope aa : % of aa in the avg. composition of a protein

↓

Assigns an antigenicity value for each amino acid from the relative occurrence of the amino acid in epitope

High antigencity value ↓ Extend to 11-13 aa

Low antigencity value ↓ Report Probable

B-Cell Epitope Prediction

Parker & Hodges method

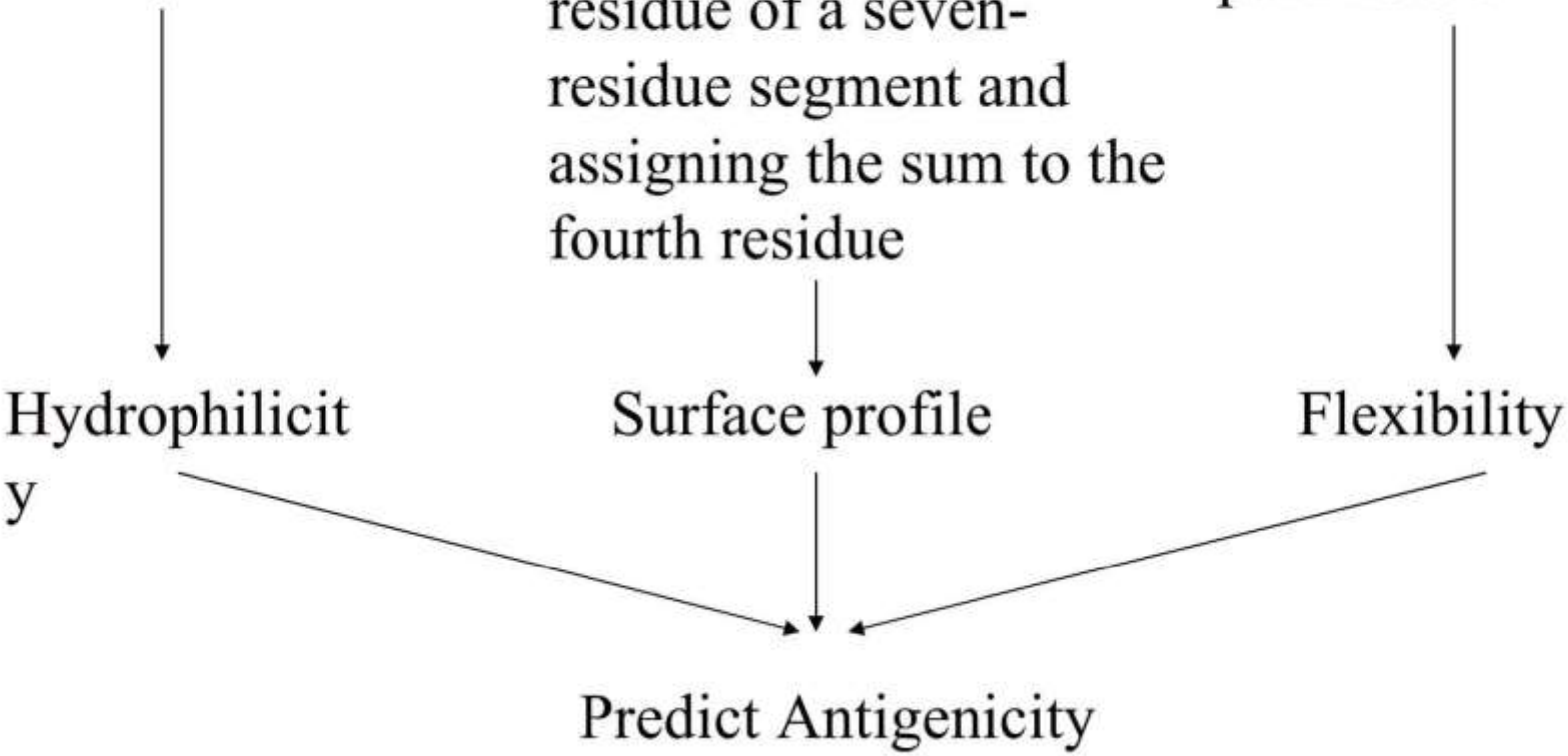
HPLC from retention co-efficient of model synthetic peptides

Janin's scale

Summing the ASA parameters for each residue of a seven-residue segment and assigning the sum to the fourth residue

Karplus & Schultz

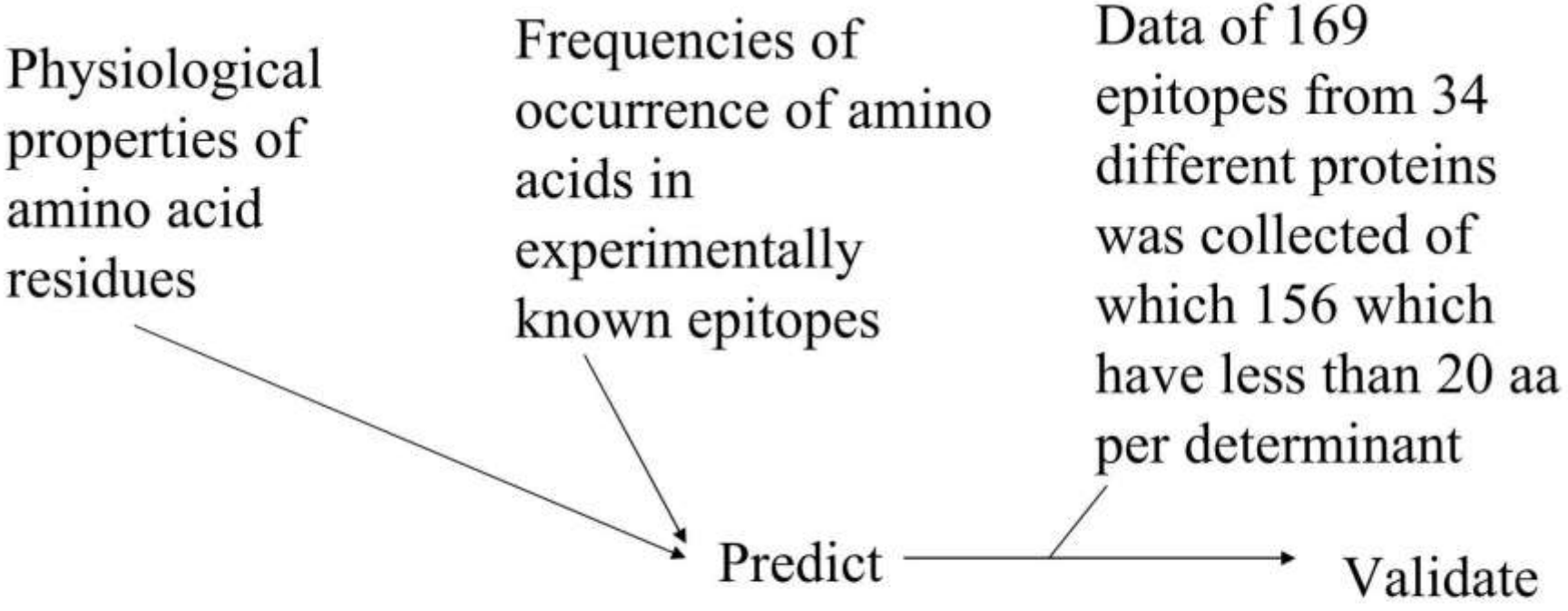
Structural parameters



B-Cell Epitope Prediction

Kolaskar & Tongaonkar's method

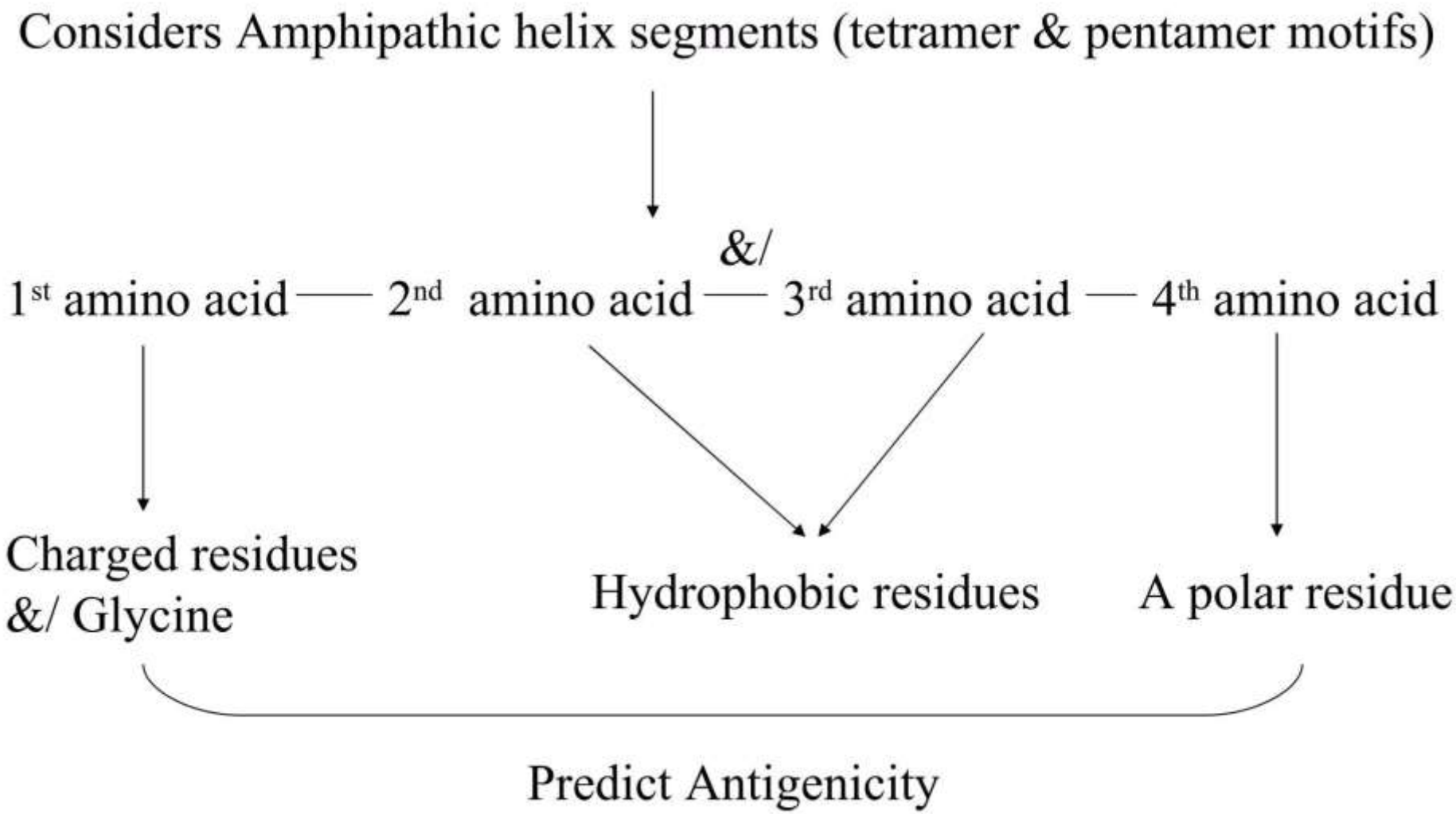
Semi-empirical method



It is available as Antigen under ExPaSy and in EMBOSS Suite

T-Cell Epitope Prediction

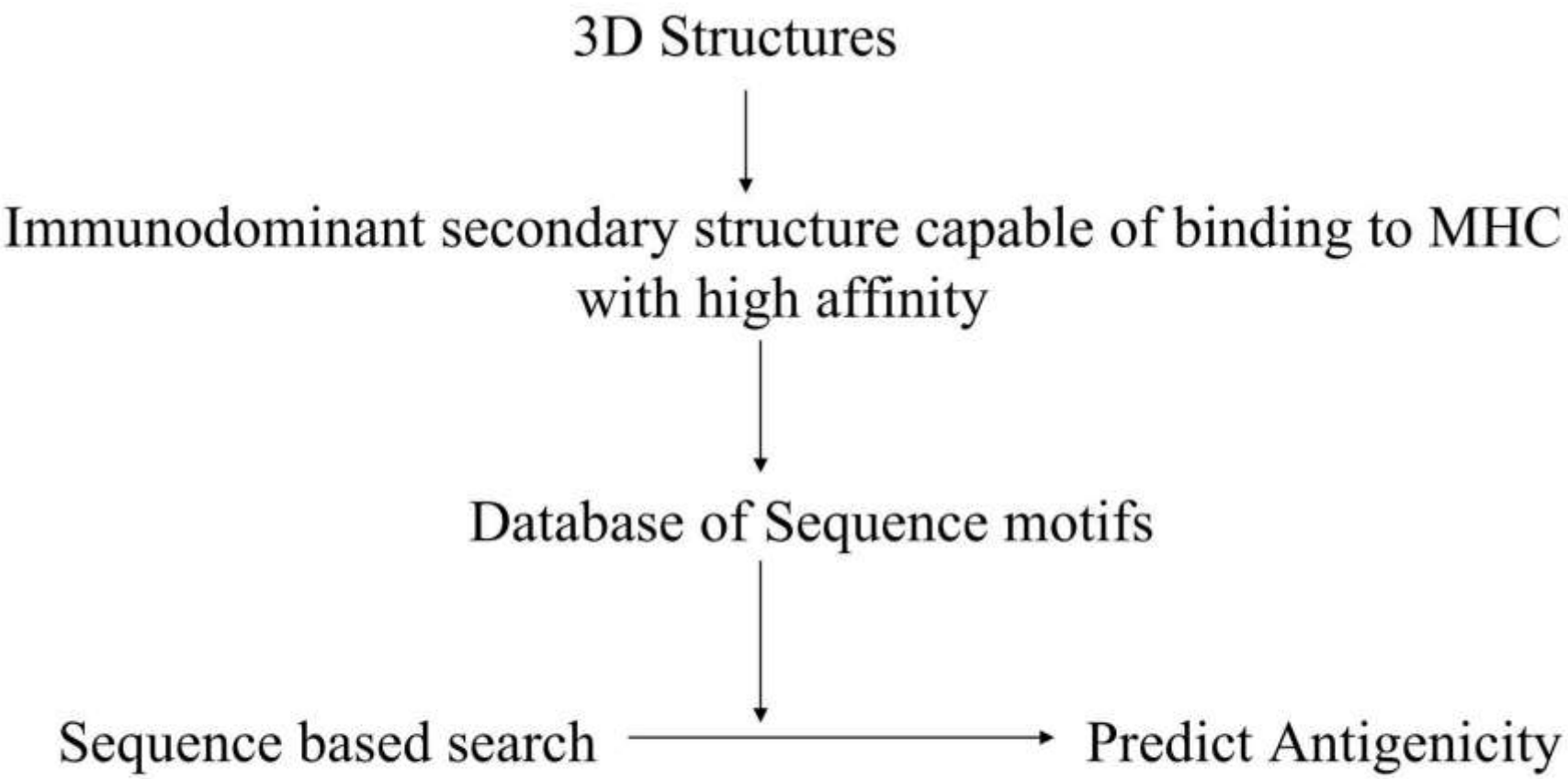
Margalit, Spouge et al. method



T-Cell Epitope Prediction

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Rothbard & Taylor method



T-Cell Epitope Prediction

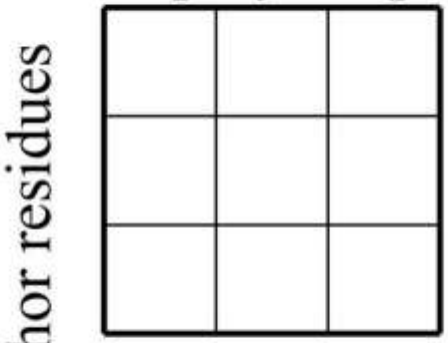
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Stille et al. method

Known MHC polymorphisms from HLA → Identify anchor residues for different polymorphisms

Construct Virtual matrices

MHC polymorphism →



Sequence based search with MHC polymorphism

Predict Antigenicity

T-Cell Epitope Prediction

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Darren R Flower et al. method

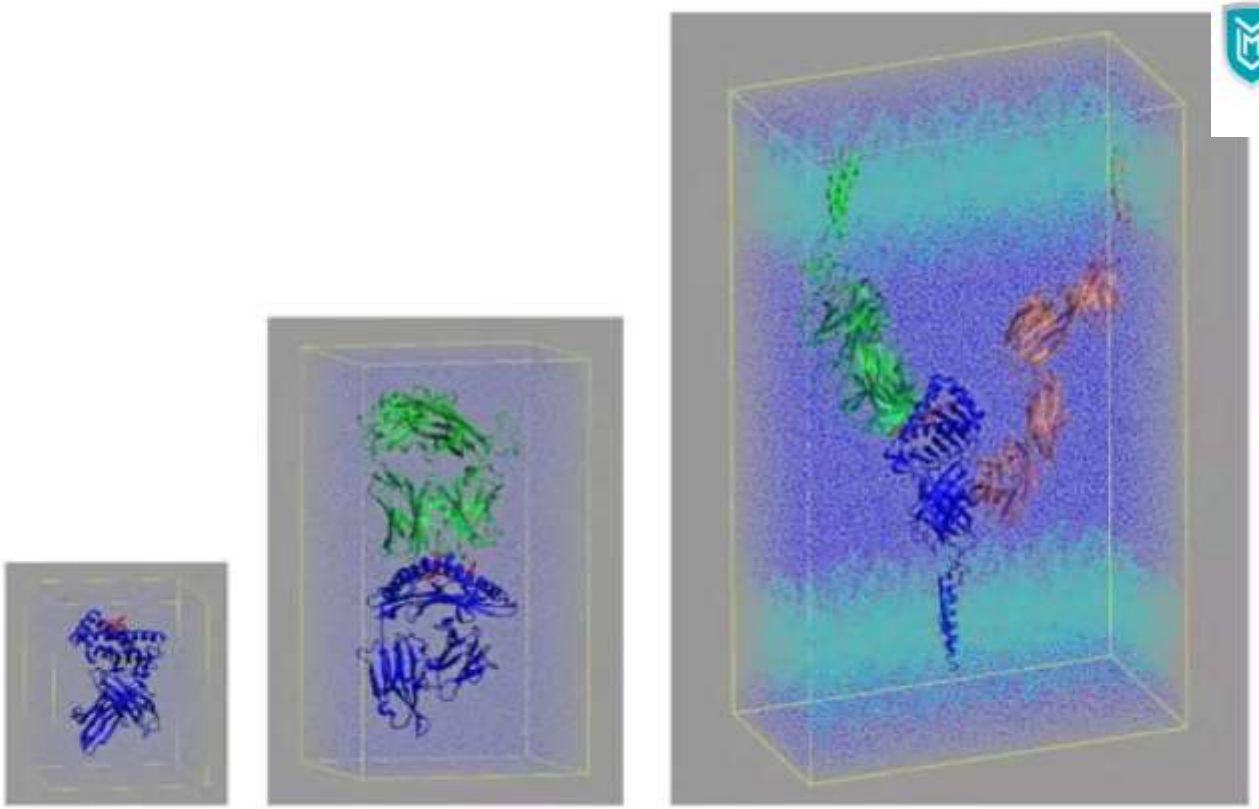
MHC binding is based on molecular dynamic simulation

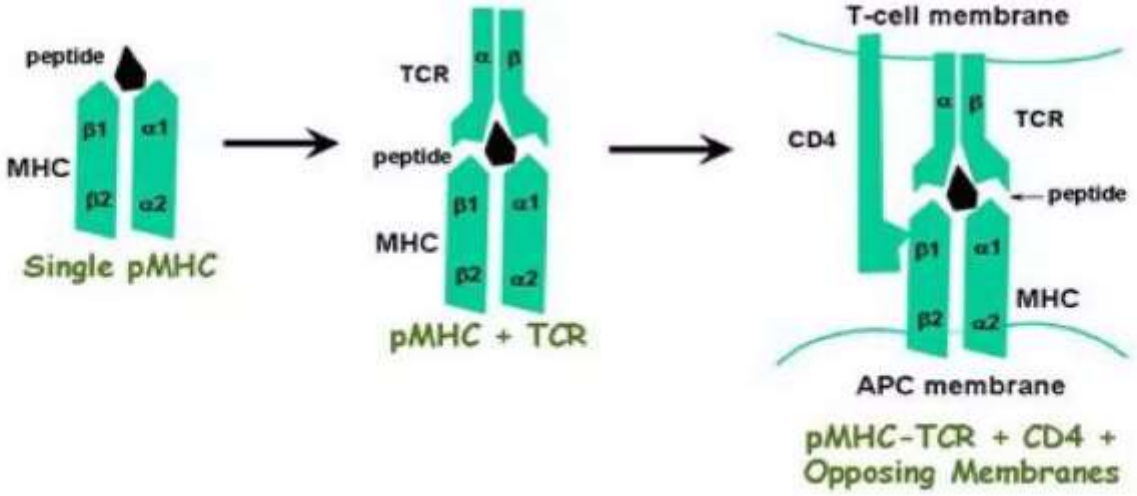
Calculate the free energy of binding for a given molecular system

No reliance on known binding data, but based on de novo prediction

Required Experimentally determined structure, or a homology model, of a MHC peptide complex

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Databases & Prediction

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MHC-binding peptides databases

SYFPEITHI
MHCPEP
JenPep
FIMM
MHCBN
HLALigand/Motif database
HIV Molecular Immunology database
EPIMHC

Prediction of MHC binding

BIMAS
SYFPEITHI
PREDEPP
Epipredict
Predict
Propred
MHCPred
NetMHC

SYFPEITHI Epitope Prediction

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Epitope prediction

This page allows you to find out the ligation strength to a defined HLA type for a sequence of aminoacids. The algorithmus used are based on the book "MHC Ligands and Peptide Motifs" by H.G.Rammensee, J.Bachmann and S.Stevanovic. The probability of being processed and presented is given in order to predict T-cell epitopes.

1. Select MHC type

If you chose "all", max. sequence length is 100 aminoacids (letters)!

- all
- H2-Ad
- H2-Ak
- H2-Db
- H2-Ed

Hold down ctrl key when clicking to select multiple items

2. Choose a mer

- octamers (8 aa)
- nonamers (9 aa)
- decamers (10 aa)
- endecamers (11 aa)
- 15 - mers (15 aa) for MHC Type II only
- all mers

3. Paste your sequence here:

Max. input 2048 aminoacids (letters)!
Letters only, no numbers or non-ASCII-symbols please.
You may use 'SYFPEITHI' with H2-Kd to see an example.

```
SYFPEITHI
```

4. Choose Run to start analysis

Run | Reset | Home

SYFPEITHI Epitope Prediction

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Search Report

Return to search conditions

BoLA-A11 decamers
BoLA-A11 octamers
BoLA-A11 nonamers
BoLA-DRB3*2703 nonamers
BoLA-HD1 nonamers
Chicken B-F12 nonamers
H2-Ab nonamers
H2-Ag7 decamers
H2-Ag7 nonamers
H2-Ak decamers
H2-Ak nonamers
H2-As nonamers
H2-Db nonamers
H2-Db decamers
H2-Kb octamers
H2-Kb nonamers
H2-Kd decamers
H2-Kd nonamers
H2-Kk octamers
H2-Kk nonamers
H2-Ld nonamers
HLA-A*01 nonamers
HLA-A*01 decamers
HLA-A*01 11 - mers
HLA-A*01 octamers
HLA-A*0201 nonamers
HLA-A*0201 decamers
HLA-A*0202 nonamers
HLA-A*0202 decamers
HLA-A*0203 nonamers
HLA-A*0203 decamers
HLA-A*0203 octamers
HLA-A*0204 nonamers
HLA-A*0205 decamers
HLA-A*0206 nonamers
HLA-A*0207 nonamers
HLA-A*03 nonamers
HLA-A*03 decamers
HLA-A*1101 nonamers
HLA-A*1101 11 - mers
HLA-A*1101 decamers
HLA-A*23 nonamers
HLA-A*2402 nonamers
HLA-A*2402 decamers
HLA-A*25 nonamers
HLA-A*26 nonamers
HLA-A*26 decamers
HLA-A*2601 nonamers
HLA-A*29 nonamers
HLA-A*3101 nonamers
HLA-A*6801 nonamers
HLA-A*6801 decamers
HLA-A*6802 nonamers
HLA-B*0702 nonamers
HLA-B*0702 decamers
HLA-B*08 octamers
HLA-B*08 nonamers
HLA-B*0801 nonamers

H2-Ld nonamers

go to top

Pos	1 2 3 4 5 6 7 8 9	score
1	T S A P D T R P A	11
3	A P D T R P A P G	10
5	D T R P A P G S T	3
2	S A P D T R P A P	2
4	P D T R P A P G S	1

THANK YOU

